Project Description: Credit Limit Increase Impact on Monthly Spend

This project investigates whether offering a **₹20,000 credit limit increase leads to a statistically significant rise in customers' average monthly spend**. This is a crucial business question for financial institutions seeking to understand the direct financial impact of such credit policy changes.

1. Objective and Hypothesis:
The core objective of this study is to determine if increasing a customer's credit limit influences their spending behavior. This was framed as a two-sample hypothesis test comparing the mean monthly spend of two groups:

- **Group A:** Customers who did not receive a credit limit increase.
- **Group B:** Customers who received a ₹20,000 credit limit increase.

The formal hypotheses for the test were:

- **Null Hypothesis (H0):** $\mu_1=\mu_2$ (There is no difference in the mean monthly spend between Group A and Group B. The credit limit increase has no effect on spending).
- **Alternative Hypothesis (H1):** μ_1偃$=\mu_2$ (The mean monthly spend is significantly different between Group A and Group B. The credit limit increase does affect spending). This is a two-tailed hypothesis, suggesting the increase could lead to either higher or lower spending.

2. Data Loading:
The project commenced by loading the relevant dataset containing customer spending information, categorized by their assigned group (A or B).
3. Exploratory Data Analysis (EDA):
A thorough EDA was performed to understand the characteristics of the spending data for both groups.

- **Summary Statistics by Group:**
  - **Group A (No Increase):**
    - Mean monthly spend: ₹30,054.70
    - Median monthly spend: ₹30,102.38
    - Standard Deviation: ₹7,850.03
    - Min spend: ₹4,069.86
    - Max spend: ₹60,821.85
  - **Group B (₹20,000 Increase):**
    - Mean monthly spend: ₹33,286.44
    - Median monthly spend: ₹33,256.78
    - Standard Deviation: ₹8,801.97
    - Min spend: ₹8,728.02
    - Max spend: ₹56,691.44 These statistics indicate that Group B had a higher

average and median monthly spend, along with a slightly higher variability.

- **Outlier Count:** Using the IQR (Interquartile Range) method, it was identified that Group A had 4 outliers, while Group B had 7 outliers. This suggests a small presence of extreme values in both groups.
- **Shapiro-Wilk Normality Test:** This test assessed whether the monthly spend data in each group followed a normal distribution.
    - Group A: W = 0.997, p-value = 0.4013. With a p-value > 0.05, the data for Group A is considered to be **normally distributed**.
    - Group B: W = 0.998, p-value = 0.6757. With a p-value > 0.05, the data for Group B is also considered to be **normally distributed**. The normality of the data for both groups is a positive finding, as it supports the use of parametric tests like the t-test.

4. Power Analysis (A Priori):
A power analysis was conducted to determine the necessary sample size to detect various effect sizes with adequate statistical power (typically 0.8) and a significance level (α) of 0.05.

- For an Effect Size (Cohen's d) of 0.2 (small effect), approximately **394 users per group** would be required.
- For an Effect Size of 0.3 (small-medium effect), approximately **176 users per group** would be required.
- For an Effect Size of 0.5 (medium effect), approximately **64 users per group** would be required.
- For an Effect Size of 0.8 (large effect), approximately **26 users per group** would be required. This analysis guides the experimental design to ensure sufficient data collection for reliable conclusions.

5. Homogeneity of Variances (Levene's Test):
Before performing a t-test, it's critical to assess if the variances of the two groups are equal. Levene's test was performed for this purpose.

- **Levene’s test p-value:** 0.0146. Since the p-value (0.0146) is less than 0.05, it indicates that the **variances between Group A and Group B are significantly unequal**. This finding is crucial as it necessitates the use of Welch's t-test, which does not assume equal variances, over a standard Student's t-test.

6. Hypothesis Test (Welch's t-Test):
Given the normal distribution of data and unequal variances, Welch's t-test was performed to compare the mean monthly spending of the two groups.

- **T-statistic:** -6.127
- **P-value:** 0.0000 With an extremely low p-value (0.0000, much less than α=0.05), the **null hypothesis was rejected**.

7. Effect Size (Cohen's d):
To understand the practical significance of the statistically significant difference, Cohen's d effect size was calculated.

- Mean Spend A (No Increase): ₹30,054.70

- Mean Spend B (Limit Increase): ₹33,286.44
- **Cohen's d: 0.388** This value is interpreted as a **small effect size**. While statistically significant, the magnitude of the difference in spending is relatively modest.

8. Confidence Interval:
A 95% Confidence Interval for the difference in mean monthly spend (Group B - Group A) was calculated using Welch's t-test.

- Difference in Means (B - A): ₹3,231.73
- **95% CI: ₹(2,196.69, ₹4,266.77)** This interval means we are 95% confident that the true average increase in monthly spend due to the ₹20,000 credit limit increase is between ₹2,196.69 and ₹4,266.77.

Conclusions:

1. **Statistically Significant Increase in Spend:** The Welch's t-test (p-value = 0.0000) unequivocally indicates a **statistically significant difference** in the average monthly spend between customers who received a ₹20,000 credit limit increase and those who did not. We reject the null hypothesis, confirming that the credit limit increase does affect spending.
2. **Positive Difference in Means:** Customers who received the credit limit increase (Group B) spent, on average, **₹3,231.73 more per month** than those who did not (Group A). The 95% confidence interval for this difference (₹2,196.69 to ₹4,266.77) reinforces a positive and consistent increase.
3. **Small Practical Effect Size:** Despite the statistical significance, the **Cohen's d of 0.388 suggests a small effect size**. This means the increase in spending, while real, is relatively modest in practical terms. It's noticeable but might not be considered a large impact.
4. **Data Normality and Variance:** Both groups' spending data were normally distributed, but their variances were unequal, correctly leading to the use of Welch's t-test for robust results.

Business Decisions:

1. **Consider Implementing Increases:** Since a statistically significant increase in average monthly spend was observed, offering a ₹20,000 credit limit increase is likely to drive higher customer spending.
2. **Evaluate ROI Against Effect Size:** While the increase is significant, the "small" effect size (Cohen's d = 0.388) implies that the **absolute increase in spending (approx. ₹3,231.73 per month)** should be carefully weighed against the associated risks (e.g., increased credit risk, operational costs) and the profit margins on the increased spend.
3. **Targeted Increases:** Further analysis could explore if certain customer segments (e.g., based on loyalty, credit score, or past spending patterns) exhibit a larger effect size. Implementing targeted credit limit increases to high-potential segments might yield a

more substantial ROI.

4. **Monitor Risk Metrics:** Alongside increased spend, closely monitor risk metrics (e.g., default rates, utilization rates) for customers receiving limit increases to ensure the strategy remains profitable and sustainable.
5. **Test Different Increase Amounts:** If feasible, consider A/B testing different credit limit increase amounts (e.g., ₹10,000, ₹30,000) to identify an optimal point that maximizes spend while managing risk.
6. **Understand Spending Drivers:** Investigate the types of purchases contributing to the increased spend. This could reveal opportunities for cross-selling or up-selling campaigns.